

Enterprise VPN Technologies: A Comparative Survey of GRE, MPLS, and IPSec Solutions

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Abstract—Virtual Private Networks (VPNs) have become a foundational component of enterprise communication infrastructure, providing secure and reliable connectivity across geographically distributed sites. This paper presents a comprehensive literature review of three widely deployed enterprise VPN technologies: Generic Routing Encapsulation (GRE), Multi-Protocol Label Switching (MPLS), and Internet Protocol Security (IPSec). We examine the architectural principles, security properties, and deployment characteristics of each technology, along with hybrid combinations such as GRE over IPSec. By analyzing recent research and practical implementations, we identify the strengths and limitations of each approach and provide a comparative discussion to guide enterprises in selecting the most appropriate VPN solution for their specific requirements. The findings suggest that no single technology is universally superior; rather, the choice depends on factors including security requirements, scalability needs, quality of service demands, and cost constraints.

Index Terms—VPN, GRE, IPSec, MPLS, enterprise network, tunneling, network security, L3VPN

I. INTRODUCTION

In the modern enterprise environment, organizations are no longer confined to a single physical location. Branch offices, remote workers, and partner organizations span cities, countries, and even continents. This geographic dispersion creates a fundamental challenge: how do we connect these sites securely and efficiently over public network infrastructure, without exposing sensitive data to unauthorized parties?

Virtual Private Networks offer a practical answer to this challenge. A VPN uses a shared or public infrastructure, such as the Internet, to create a private communication channel that mimics the behavior of a dedicated physical network [1]. From the perspective of the end user, traffic appears to flow through a private pipe, even though it physically travels through shared infrastructure. VPNs provide the confidentiality, integrity, and availability properties that enterprise operations demand.

Three technologies dominate enterprise VPN deployments today. The first is Generic Routing Encapsulation (GRE), a tunneling protocol defined in RFC 2784 [2] that encapsulates one network protocol inside another. GRE is lightweight and flexible but provides no native encryption. The second is IPSec, a comprehensive security framework defined in RFC 4301 [3] that provides authentication and encryption at the network layer. IPSec is the de facto standard for securing

VPN traffic over the Internet. The third is MPLS, defined in RFC 3031 [4], a high-performance packet forwarding mechanism that enables service providers to deliver scalable, multi-tenant VPN services with strong traffic isolation and Quality of Service (QoS) guarantees.

Each technology has found a distinct place in enterprise networking. GRE is commonly used for its simplicity and routing flexibility. IPSec is deployed wherever strong encryption is required. MPLS is the preferred solution for service providers and large enterprises that need scalability and traffic engineering. In many real-world deployments, these technologies are combined — most notably in the form of GRE over IPSec — to leverage the strengths of more than one approach simultaneously.

This paper surveys the literature on all three technologies and their combinations. The goal is to provide a clear, structured understanding of how each technology works, what it is good for, and where it falls short. The rest of the paper is organized in such a way that Section III discusses related work found in the literature, followed by a discussion in Section IV and the paper is concluded in Section V.

II. BACKGROUND

Before reviewing specific technologies, it is useful to briefly establish the general VPN taxonomy that the literature commonly uses. VPNs can be classified along several dimensions. By responsibility, they are either customer-provisioned (the customer manages the VPN endpoints) or provider-provisioned (the service provider manages the VPN infrastructure) [5]. By layer of operation, they are either Layer-2 VPNs, which operate at the data-link layer and present a virtual switch to the customer, or Layer-3 VPNs, which operate at the network layer and involve the service provider in IP routing [6]. By access model, they are either site-to-site VPNs, which connect fixed locations, or remote access VPNs, which serve mobile and home users [1].

The three technologies covered in this paper map to different positions in this taxonomy. GRE is a customer-side tunneling mechanism that operates at Layer-3. IPSec is a security framework that operates at Layer-3 and is used in both customer and provider deployments. MPLS is primarily a provider-side

technology that supports both Layer-2 and Layer-3 VPNs at significant scale.

III. RELATED WORK

A. Tunnel-Based VPN Technologies

1) *Generic Routing Encapsulation (GRE)*: GRE is one of the oldest and most widely used tunneling protocols in enterprise networking. It was standardized by Cisco and later published by the IETF in RFC 2784 [2]. The fundamental idea behind GRE is simple: a packet of one protocol, such as IPv6, is encapsulated inside a GRE header and then placed inside an outer IPv4 packet for delivery across a network that would not otherwise support the inner protocol. At the receiving end, the outer header and GRE header are removed and the original packet is forwarded normally.

GRE tunnels are stateless, meaning the tunnel itself does not maintain session information. This makes GRE lightweight and easy to configure. A common real-world use case, demonstrated in the work of Fahriza et al. [7], is IPv6-over-IPv4 tunneling, where an organization needs to deploy IPv6 across branches but the WAN infrastructure only supports IPv4. GRE handles this encapsulation cleanly and enables OSPF routing to run end-to-end across the tunnel, ensuring efficient convergence.

However, GRE has one significant limitation: it provides no encryption and no authentication on its own. A GRE tunnel passes traffic as plaintext, meaning any attacker who can observe traffic on the underlying network can read the contents of the tunnel. GRE was never intended as a security mechanism; it is a routing and encapsulation tool.

Uddin et al. [8] conducted an analysis of GRE over IPv6 networks and confirmed that while GRE provides the routing flexibility needed to connect heterogeneous networks, the absence of native security makes it unsuitable as a standalone enterprise VPN solution where data confidentiality is required. Their work reinforces the view that GRE is best understood as a building block to be combined with security protocols, rather than a complete VPN solution in itself.

Ogudo [9] examined GRE and IPsec tunneling protocols specifically for secure communication over public networks and concluded that GRE alone is appropriate only in trusted or internally controlled environments. For public Internet traversal, the combination of GRE with IPsec is necessary.

2) *IPsec Protocol Suite*: IPsec is a comprehensive framework defined across multiple RFCs, with the architecture specified in RFC 4301 [3]. Unlike GRE, IPsec was designed from the ground up as a security solution. It operates at the IP layer and is transparent to applications above it, which means no application modification is required for IPsec to protect traffic.

IPsec provides three core security services. The first is confidentiality, achieved through encryption using algorithms such as AES. The second is data integrity, achieved through keyed hash functions that verify packets have not been tampered with in transit. The third is authentication, which verifies the identity of communicating parties. These services are delivered

through two protocols: the Authentication Header (AH), which provides integrity and authentication but does not encrypt the payload, and the Encapsulating Security Payload (ESP), which provides all three services including confidentiality, making it the more commonly deployed option in practice.

IPsec operates in two modes. In transport mode, only the payload of the IP packet is protected and the original IP header is preserved. In tunnel mode, the entire original IP packet is encapsulated inside a new IP packet and fully protected. Tunnel mode is the standard choice for gateway-to-gateway VPNs, where the VPN endpoints are different from the actual source and destination hosts [1].

Key management in IPsec is handled by the Internet Key Exchange (IKE) protocol. IKE version 1 was widely deployed but has known vulnerabilities. IKEv2 resolves these issues and offers improved authentication and negotiation efficiency. Most modern deployments use IKEv2.

Nair [10] provided a detailed examination of IPsec integration for data security and concluded that IPsec with ESP in tunnel mode is the most robust option for enterprise site-to-site VPNs. The combination of AES-256 for encryption and SHA-256 for integrity provides strong protection against both passive eavesdropping and active tampering attacks.

Rosu et al. [1] studied a large geographically dispersed enterprise and implemented an IPsec-based VPN architecture connecting a headquarters in Bucharest with hundreds of branch offices across Romania. Their implementation used IPsec tunnel mode with DES and 3DES encryption deployed on Cisco routers. The authors found that IPsec met the confidentiality and integrity requirements of the organization and integrated cleanly with the existing routing infrastructure. The deployment also demonstrated that open-source network monitoring tools like Nagios can effectively manage the health of large-scale IPsec VPN networks.

Sankaran [11] examined cross-protocol attacks on IPsec-based VPNs and identified several attack vectors including IKEv1 downgrade attacks and traffic analysis. This work highlights that while the IPsec cryptographic core is strong, the overall security depends on careful configuration, particularly in the choice of IKE version and cipher suites. The author recommends mandatory use of IKEv2 and avoidance of legacy cipher suites such as 3DES in any new deployment.

3) *GRE over IPsec (Hybrid Approach)*: Because GRE lacks security and IPsec lacks routing flexibility, combining the two has become a standard practice in enterprise networking. In a GRE-over-IPsec configuration, a GRE tunnel is first established between two routers, and then IPsec is applied to encrypt all traffic flowing through that tunnel. The result is a tunnel that supports dynamic routing protocols like OSPF and EIGRP — which cannot run directly over a raw IPsec tunnel — while also providing full encryption of all encapsulated traffic.

Fahriza et al. [7] implemented exactly this architecture for a private network simulation in Indonesia. Their topology consisted of three regional branch offices connected through GRE tunnels with IPsec encryption, using OSPF for routing and

ISAKMP for IPsec key management. Network performance was evaluated by measuring QoS metrics including delay, jitter, and throughput. The simulation results showed that the GRE-IPsec combination successfully established secure connections between all branch sites, and network quality remained acceptable even when tunneling was active. The authors also conducted a techno-economic analysis and found the implementation financially viable, with an Internal Rate of Return of 84.80% and a payback period of approximately 11 months.

Kumar et al. [12] also studied IPsec with encapsulation and found that the combination of GRE and IPsec, when properly configured, provides an effective balance between routing functionality and security. Their work specifically examined how multicast and broadcast traffic, which IPsec cannot carry directly, can be supported through GRE encapsulation before IPsec encryption.

The main tradeoff in GRE-over-IPsec is overhead. Each packet carries both a GRE header (4 bytes minimum) and IPsec headers. This additional overhead reduces the effective MTU of the tunnel and increases CPU load on the router. For environments with high throughput requirements or latency-sensitive applications, this overhead must be accounted for in capacity planning.

B. Provider-Managed VPN Technologies: MPLS

1) *MPLS Architecture and Core Concepts:* Multi-Protocol Label Switching was developed by the IETF to address the limitations of traditional hop-by-hop IP routing. MPLS is defined in RFC 3031 [4] and introduces a 32-bit label inserted between the Layer-2 header and the Layer-3 IP header. Routers in an MPLS domain — called Label Switching Routers (LSRs) — forward packets based on this label rather than performing a full IP routing table lookup at each hop. This label-based forwarding is faster and more predictable than conventional IP routing.

The MPLS label structure consists of a 20-bit label value, 3 bits for experimental (EXP) use, 1 bit for bottom-of-stack indication, and 8 bits for Time-to-Live (TTL). Labels are assigned and distributed among routers using the Label Distribution Protocol (LDP) or the Resource Reservation Protocol with Traffic Engineering extensions (RSVP-TE).

At the edges of the MPLS domain, Provider Edge (PE) routers receive traffic from Customer Edge (CE) routers and assign labels before forwarding into the MPLS core. Provider (P) routers in the core only perform label switching and have no knowledge of customer VPN information. At the far edge, the receiving PE router removes the label and delivers traffic to the destination CE router. This penultimate hop popping mechanism, where the label is removed one hop before the destination, further reduces forwarding overhead.

Mungse [13] provides a detailed examination of this architecture, including the role of Virtual Routing and Forwarding (VRF) tables in maintaining traffic isolation between different customers. Each VRF on a PE router acts as a completely separate routing table for a specific customer VPN. This

isolation ensures that routes and traffic from one customer are never visible to another, even though both share the same physical infrastructure.

2) *Layer-2 MPLS VPNs:* Layer-2 MPLS VPNs present the service provider network as a virtual switch to the customer. The customer sees nothing but Layer-2 circuits between their sites, and the provider handles all internal routing and label switching. Two main variants exist. Virtual Private Wire Service (VPWS) provides point-to-point Layer-2 circuits. Virtual Private LAN Service (VPLS) provides a multi-point Layer-2 service that appears to the customer as a single, shared LAN switch.

Fayed et al. [6] note that in a Layer-2 VPN, the service provider network is transparent to the customer at Layer-3. This means the customer is fully responsible for their own IP addressing and routing, and point-to-point addresses must be allocated at both customer sites from the same subnet. While this simplicity gives customers full routing control, it also means that the provider cannot optimize traffic paths based on Layer-3 information, which can limit efficiency in complex topologies.

3) *Layer-3 MPLS VPNs (BGP/MPLS VPNs):* Layer-3 MPLS VPNs, defined in RFC 4364 [14], are the most widely deployed MPLS VPN type in enterprise and carrier environments. In an L3VPN, the service provider participates directly in the customer's routing by maintaining separate VRF instances for each customer on the PE routers. BGP with Multiprotocol extensions (MP-BGP) distributes VPN routing information between PE routers.

Two key concepts make L3VPNs function correctly. The first is the Route Distinguisher (RD), a 64-bit value prepended to each customer route that makes it globally unique, even if multiple customers use overlapping IP address spaces. The second is the Route Target (RT), a BGP community attribute that controls which routes are exported from and imported into each VRF. By carefully configuring RTs, service providers can implement complex connectivity models such as full mesh, hub-and-spoke, and overlapping VPNs.

Mungse [13] implemented a complete L3VPN in GNS3 with PE, P, and CE routers. The implementation demonstrated VRF-based traffic isolation, BGP route distribution, and multi-site reachability. Ping and traceroute tests confirmed that Site A traffic reached all designated Site B destinations while remaining completely isolated from other customers' traffic. Wireshark captures confirmed BGP UPDATE messages carrying VPN route information between PE routers.

Fayed et al. [6] extended this work to examine more complex MPLS VPN scenarios. They implemented and validated three advanced models on a live network emulation using GNS3 with Cisco IOU software. The first was the simple VPN, where each customer's sites communicate only among themselves. The second was the overlapping VPN, where customer Headquarters sites communicate with each other across VPN boundaries while branch sites remain isolated. This is achieved by assigning additional Route Targets to the HQ PE routers, enabling controlled route leaking between otherwise

separate VPNs. The third was the central services VPN, where a shared server farm is accessible to all customers through their individual VPNs while customers remain isolated from each other. All three models were validated by inspecting BGP routing tables of customer routers and confirming expected route presence and absence.

The central services model is particularly relevant for enterprise deployments, as it allows a shared security infrastructure, Active Directory, or DNS service to serve multiple customer segments without requiring those segments to communicate directly with each other.

Huda and Andriyani [15] further demonstrated how GRE tunnels can be used in conjunction with OSPF multi-area designs, showing that GRE and MPLS need not be viewed as competing technologies but can complement each other in hybrid WAN architectures.

IV. DISCUSSION

A. Security Comparison

The three technologies differ fundamentally in their approach to security. IPsec provides the strongest built-in security, offering encryption, integrity verification, and mutual authentication in a single framework. GRE alone provides no security whatsoever. MPLS VPNs rely on traffic isolation rather than encryption; customer traffic is segregated at the routing layer through VRF tables and label switching, but the traffic itself is not encrypted in transit.

This distinction matters in practice. An MPLS VPN deployed over a service provider's managed infrastructure is generally considered secure because the provider controls the core network and prevents unauthorized access. However, MPLS VPN traffic running over a shared or untrusted backbone provides no protection for the data. In such scenarios, IPsec must be added as an overlay to provide encryption [6]. For organizations that must comply with data protection regulations, IPsec or GRE-over-IPsec is the appropriate choice, as it provides end-to-end encryption regardless of the underlying transport.

B. Scalability and Management

MPLS VPNs have the strongest scalability story of the three technologies. The VRF-based architecture separates customer routing tables at PE routers, meaning P routers in the core carry no VPN-specific state. Adding a new customer VPN requires configuration only at the edge. This property allows service providers to support thousands of enterprise customers on the same physical infrastructure without increasing the complexity of the core [13].

GRE tunnels do not scale well in full-mesh deployments. If an enterprise has N sites and each must reach all others, $N(N-1)/2$ tunnel pairs must be configured. For even moderately sized networks this becomes administratively burdensome. IPsec suffers from similar scaling limitations, as each tunnel requires its own Security Association (SA). MPLS also excels in supporting complex connectivity models through the RT/RD mechanism, which allows fine-grained control over

site reachability — including overlapping and central services models — with no equivalent in GRE or IPsec [6].

C. Quality of Service

MPLS supports QoS through the EXP bits in the label header, which mark different traffic classes for differential treatment within the provider network. This allows a service provider to guarantee bandwidth, delay, and jitter for specific traffic classes, critical for voice and video applications. Service Level Agreements for delay-sensitive traffic are routinely backed by MPLS traffic engineering in carrier networks [6].

GRE and IPsec do not inherently support QoS. When traffic is encapsulated in a GRE or IPsec tunnel, inner DSCP markings may or may not be preserved in the outer header, depending on configuration. In practice, QoS over GRE-IPsec tunnels requires careful end-to-end planning and is much harder to guarantee than in a managed MPLS environment.

D. Cost and Deployment Context

GRE and IPsec are software-based features available on virtually all modern enterprise routers and firewalls, often at no additional cost. They can be deployed quickly over any IP network, including commodity Internet connections. For small to medium enterprises with limited budgets, GRE-over-IPsec over broadband Internet is a very cost-effective VPN solution.

MPLS VPNs typically require a service provider subscription. However, for large enterprises with stringent SLA requirements, the cost is often justified by the reliability, QoS guarantees, and reduced management overhead. Fahriza et al. [7] demonstrated through detailed financial modeling that even a self-managed GRE-IPsec deployment can be economically favorable, showing a positive Net Present Value, an IRR of 84.80%, and a payback period of under one year for a three-branch Indonesian enterprise.

E. Summary Comparison

Table I summarizes the key differences across the four configurations discussed in this paper.

TABLE I
COMPARATIVE SUMMARY OF ENTERPRISE VPN TECHNOLOGIES

Criterion	GRE	IPSec	GRE/IPSec	MPLS L3VPN
Encryption	None	Strong	Strong	None*
Authentication	None	Strong	Strong	Implicit
Routing Support	Full	Limited	Full	Full (BGP)
QoS Support	Basic	Basic	Basic	Native
Scalability	Low	Low	Low	High
Configuration	Low	Medium	Medium	High
Cost	Low	Low	Low	High
Multicast	Yes	No	Yes	Yes
Deployment Side	Customer	Customer	Customer	Provider

*MPLS provides traffic isolation, not encryption.

The table shows that no single technology dominates across all criteria. GRE-over-IPsec is the most practical choice for enterprises that manage their own WAN connectivity over

the Internet. MPLS L3VPN is the preferred choice for enterprises that require carrier-grade QoS, scalability, and complex connectivity models, and are willing to pay for a managed service. Organizations with very high security requirements may choose to run IPsec over their MPLS circuits as an additional protection layer against insider threats at the service provider.

V. CONCLUSION

This paper reviewed three foundational enterprise VPN technologies — GRE, IPsec, and MPLS — from the perspective of their technical operation, security properties, and practical deployment characteristics. The review covered tunnel-based approaches including standalone GRE, IPsec in tunnel mode, and the widely deployed combination of GRE-over-IPsec, as well as provider-managed MPLS VPNs including both Layer-2 and Layer-3 variants with complex deployment models.

Several conclusions emerge from this survey. First, GRE should never be deployed alone in environments where data confidentiality is required. Its value lies in routing flexibility, and it is best used as an inner tunnel encapsulated within IPsec. Second, IPsec remains the gold standard for encrypting VPN traffic, and IKEv2 with modern cipher suites such as AES-256 and SHA-256 should be the mandatory configuration in any new deployment. Third, MPLS L3VPNs offer capabilities in scalability, QoS, and flexible connectivity that GRE and IPsec simply cannot match. For large enterprise and service provider environments, MPLS with BGP-based L3VPN remains the most powerful solution available.

Future work should examine the convergence of traditional VPN technologies with Software-Defined Networking (SDN) and cloud-native architectures. As enterprises increasingly move workloads to the cloud, the boundaries between GRE, IPsec, MPLS, and cloud-native VPN solutions are blurring, and new hybrid architectures are emerging that combine elements of all three.

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